

EK 210 A7 Group 7 - Line Following Car Report

1. Executive Summary

The goal of this line following car is to fulfill the objective of traversing a path, indicated by a black line approximately 1 inch in width, while carrying a 250 gram mass, stopping when faced with temporary obstacles or the finish line (a perpendicular black line 4 inches in width affixed to the end of the standard path), and following lengths of path with both sharp and curved turns of no more than 135 degrees. This car is designed to perform this task autonomously, with the user input being activating the line-following algorithm remotely once it is at the start of the path.

In order to achieve this, the line following car employs 4 DIANN 3-6V DC motors, 2 L298 DC Motor Drivers, and 4 corresponding 1:48 gear model wheels for actuation of the vehicle, an Arduino Uno for signal processing and output, an Adafruit TSOP38238 IR Remote Receiver and Adafruit Mini IR Remote controller for wireless activation. For detection of the line and obstacles, 3 HiLetgo IR Infrared sensors were used (2 for line, 1 for obstacles). As for power, one 12V rechargeable Lithium-ion battery and one 9V battery were used to power the motors and Arduino respectively. As for housing components, the motor mounts were made from laser cut $\frac{1}{8}$ " acrylic and the rest of the vehicle's chassis was 3D printed from PLA. The car achieved all objectives of speed, size, cost, accuracy, and battery life. This project was significant for its demonstration of highly affordable embedded systems and optimization for a relatively complex task at a low economic and power budget.

2. Introduction, problem statement, key objectives and metric

To satisfy our client, our team was tasked with designing and building an autonomous, load-bearing line following car. This vehicle was to be capable of tracing a predefined black line path approximately one inch wide, while carrying a 250-gram mass. Designed for use by engineering students, the project emphasized real-world embedded control and mechatronic design principles. The car had to start autonomously with a wireless signal, navigate a path featuring straightaways, curves, and sharp turns up to 135°, and respond to obstacles or a finish line by stopping appropriately. In addition to ensuring technical performance, the design also had to stay within a budget and adhere to spatial constraints. The key performance objectives for the car included path accuracy, environmental responsiveness, cost-effectiveness, and completion speed (see Table 1 and Appendix C). Specifically, the vehicle needed to remain consistently on the line, respond to obstacles within 3 seconds, and complete the course in under 30 seconds, all while staying within a \$200 budget (Appendix C). These requirements guided both hardware and software decisions, from the choice of Arduino Uno for control, to the integration of HiLetgo IR sensors for tracking and obstacle detection (see Table 2). Through iterative testing and refinement, the final design successfully met all core metrics, offering our client a modular,

robust, and responsive platform to explore sensor integration, motor control, and autonomous navigation.

3. Important Design Elements and Rationale

The most critical feature of this design is its path-following mechanism. We considered several means of path and obstacle detection in the early design phase (See Table 2, Rows 5 and 6), with infrared sensors ultimately proving to be the most appropriate choice for the task. The device is equipped with three infrared sensors evenly spaced across the front of the body (see Figure 5). This arrangement was selected because it effectively addresses the question of how to have the car identify turns and deviations autonomously and respond in real time. The middle sensor is used for obstacle detection; if it is triggered, all four motors stop spinning. The left- and right-sensors are used to continuously assess the car's alignment on the path and the need for course-correction. Either of them being tripped will trigger the wheels on the same side to stop and those opposite to spin to incrementally pivot until the sensor no longer detects the line. This method of turning is more suitable for our purposes than individually-pivoting wheels because it allows for sharp direction changes without the need for forward motion.

4. Final Description of Product

The result is a completed modular Arduino Uno-controlled line-following car built with Wago connectors for tool-free, secure joining. Three HiLetgo IR sensors locate the track and obstacles, stopping or steering the car to the left or right as needed, and four B0BR7S1DQZ 6V DC motors, controlled by two L298 motor drivers, provide smooth, load-enabled movement (see Figure 1, 3 and 5). It runs on a high-capacity 9 V battery with an added 12 V rechargeable lithium-ion power supply to ensure almost 5 hours of uninterrupted operation (see Figure 5, Appendix D, and Eq. 1). Start/stop and mode changes are remotely controlled via an Adafruit Mini IR remote and a TSOP38238 receiver. It is mounted on a 1:48 rubber-tire track and allows students to have direct experience with embedded control, sensor fusion, and mechatronic design (see Figure 4). The entire diagram of the vehicle is shown in Figure 2.

Completed to autonomously trace a one-inch-wide black line while carrying a 250 g payload, the car pauses for temporary obstacles or when it detects the four-inch-wide finish line. It effortlessly handles straight segments and both sharp and sweeping turns up to 135°, requiring only a touchless start command at the beginning of the track (see Figure 4).

5. Evaluation of Results

Our vehicle met every objective set (see Appendix C), with the only debatable shortcoming being the distance of obstacle detection . Over the 6 trials that our vehicle was tested, there were no cases in which our vehicle took longer than 1 minute 30 seconds to complete the course, and in fact averaged around 1 minute and 18 seconds (Figure 6, Table 4), completing our speed objective. The vehicle also succeeded in operating reasonably for 1 hour

without running out of power, as after 2 full course traversals and 60 minutes the power sources of the system remained sufficient for further use (Table 5, Figure 7). For the first 2 course traversal trials, the vehicle's obstacle detection distance was below the objective 10 cm, but after adjustment of the sensor, the vehicle succeeded with averages in excess of 10 cm of distance when the vehicle stopped due to an obstacle (Table 6, Figure 8). The car remained on the line at all times, was successfully activated wirelessly, and stopped at the finish line for all trials. The car has a profile of 7.75" x 8.375" that fits within the 8.5" x 11" expected area. We also remained under the budget, with a total BoM of \$73.94 out of the \$200 in our goals (Table 3).

These results not only validated our design choices and constraints but also offered engineering students valuable hands-on learning. By participating in prototyping, sensor calibration, performance testing, and data analysis, students applied theoretical concepts to practical problems and gained experience in iterative design and problem solving. Moreover, the project demonstrated highly affordable embedded systems and optimized a relatively complex task within tight economic and power budgets.

6. Lesson Learned

In reflecting on this project, we learned that creating neat, well-organized wiring from the start can save hours of future troubleshooting, and that resisting the urge to overengineer simple systems allows us to focus on core functionality before adding unnecessary features. We discovered the importance of realistic scheduling—having sufficient buffer room for integration, debugging, and coding is critical to avoid last-minute rushes—and built in some incremental test phases so that issues could be caught and corrected early, instead of propagating to cause cascading errors later. Aesthetics can be tempting, but prioritizing performance over cosmetics enabled us to deliver a good prototype sooner. No less important was the self-control to measure twice and double-check every dimension before finalizing 3D prints, minimizing waste and rework. Finally, up-front sensor calibration, thoughtful power-budgeting planning, and clear documentation of design changes greatly improved our development productivity and teamwork.

Appendix A - Figures

Figure 1 - Glass box

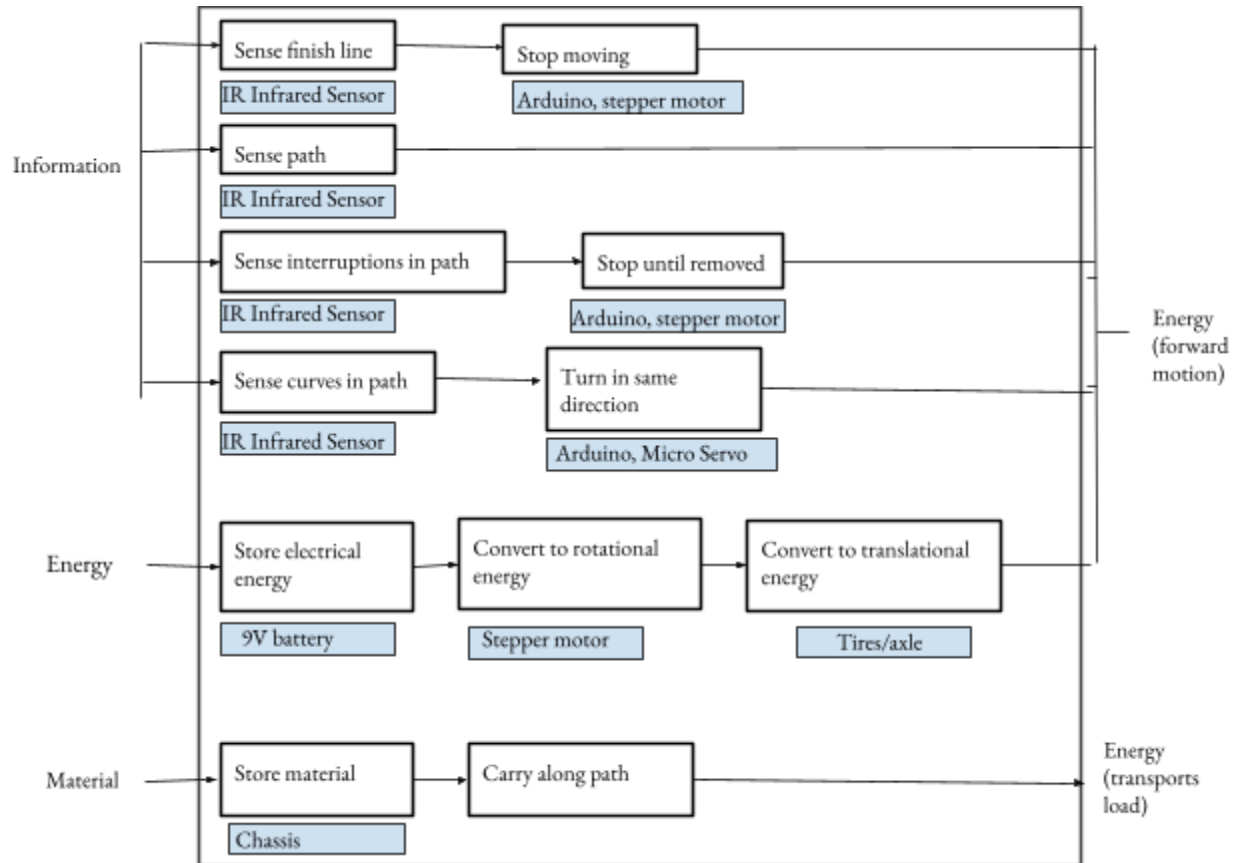


Figure 2 - Electronics Sketch

(https://docs.google.com/drawings/d/1QLb3mcTyyYUKRiz9cb2GB58_JGE08yRmoMI9dVUipS4/edit)

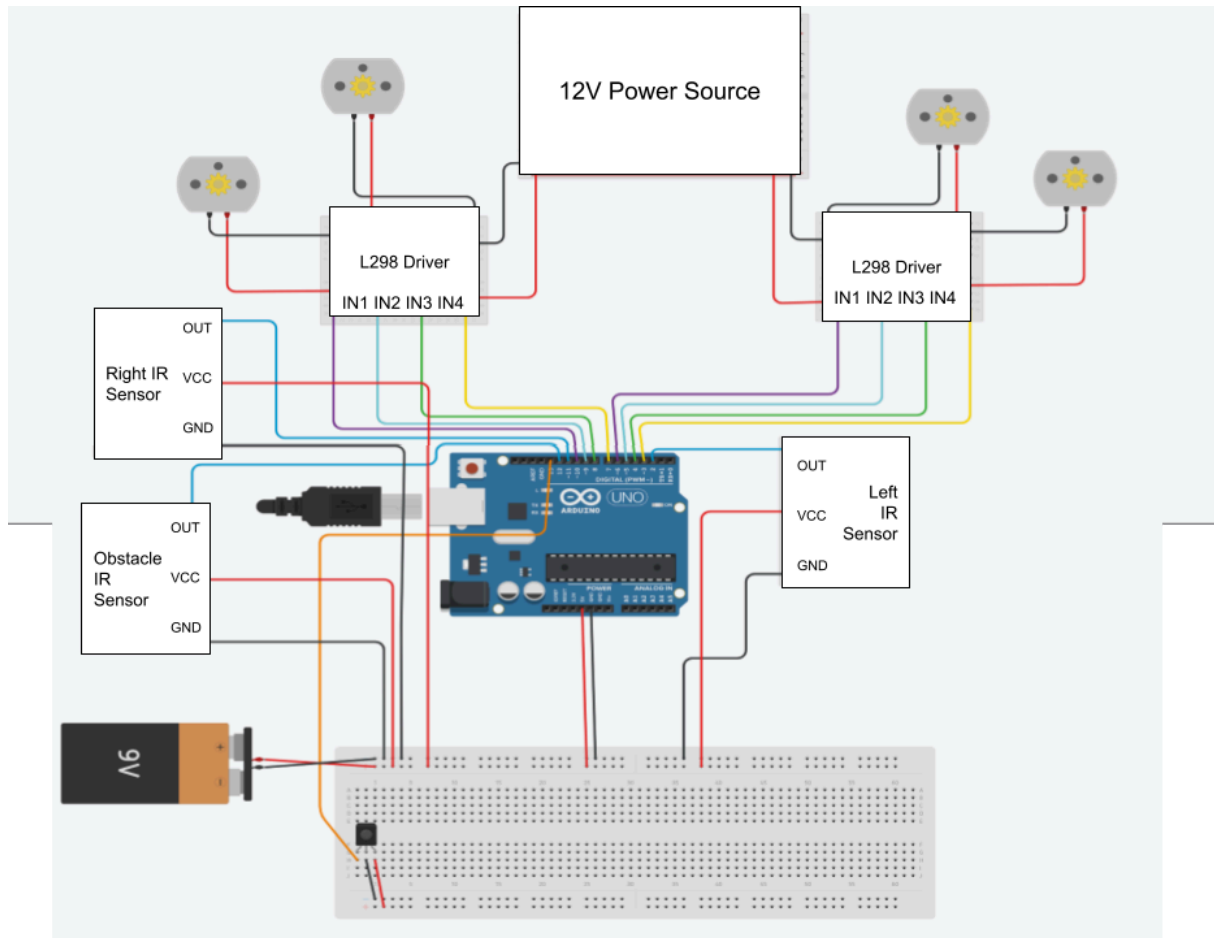
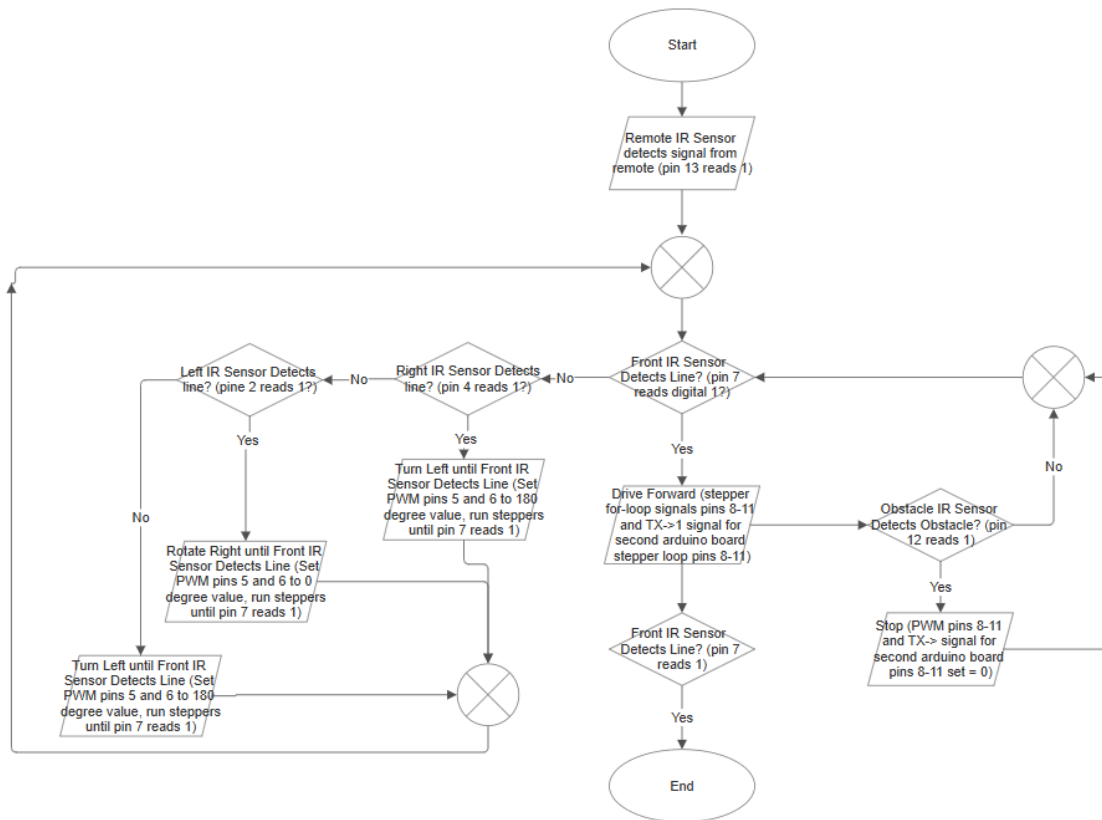


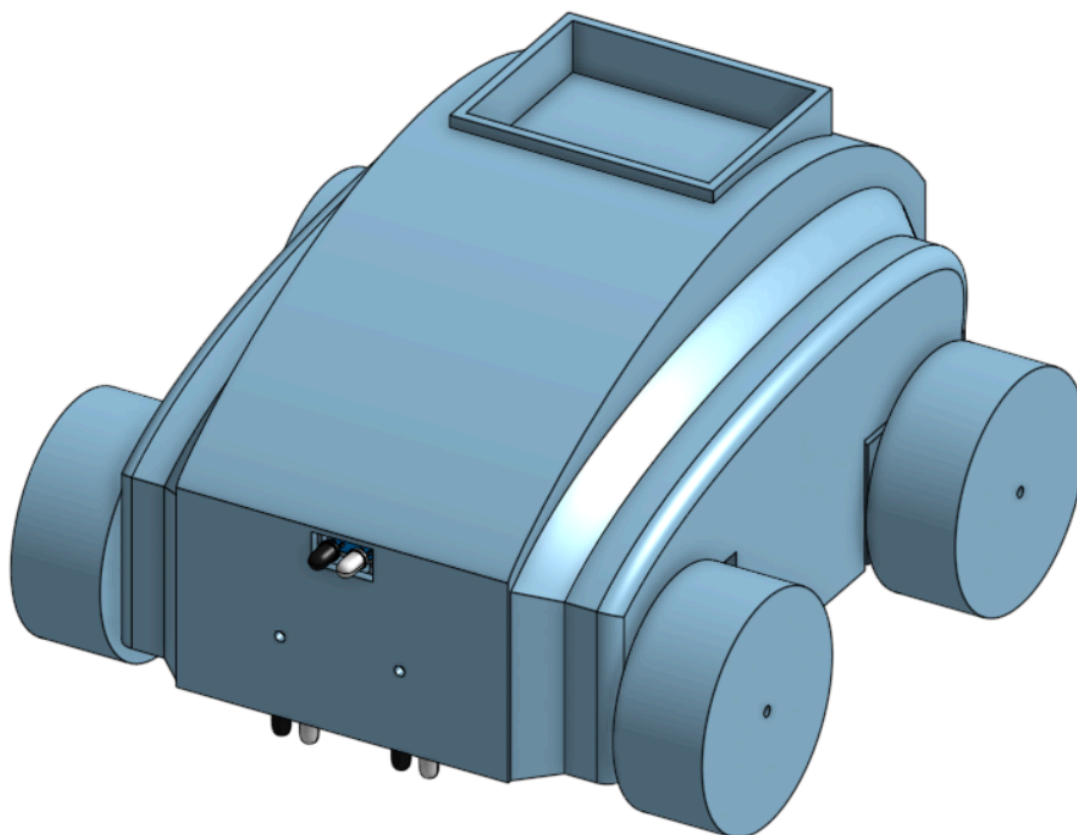
Figure 3 - Code Flow Chart



Figures 4 and 5 - CAD

(<https://bu.onshape.com/documents/4798bb8d54f4daaa8a8febab/w/858ae3854ec6090c93df9789/e/eb606d2dec0dca850e01093>)

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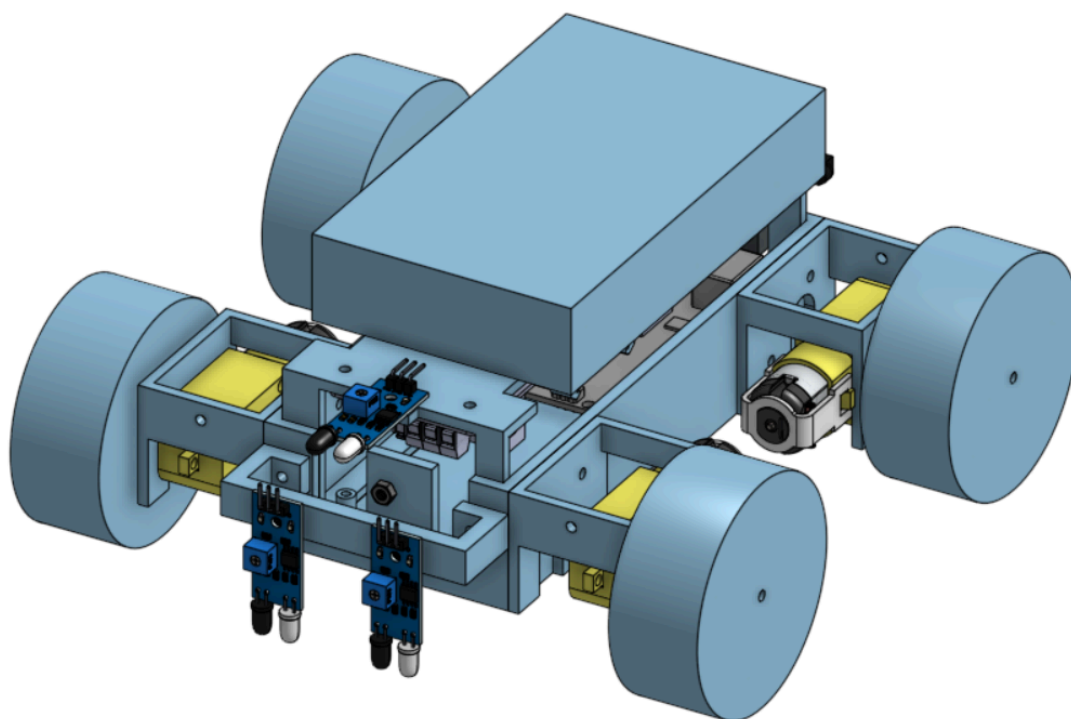


Figure 6 - Course Time Trial Graph

Course Time Trials

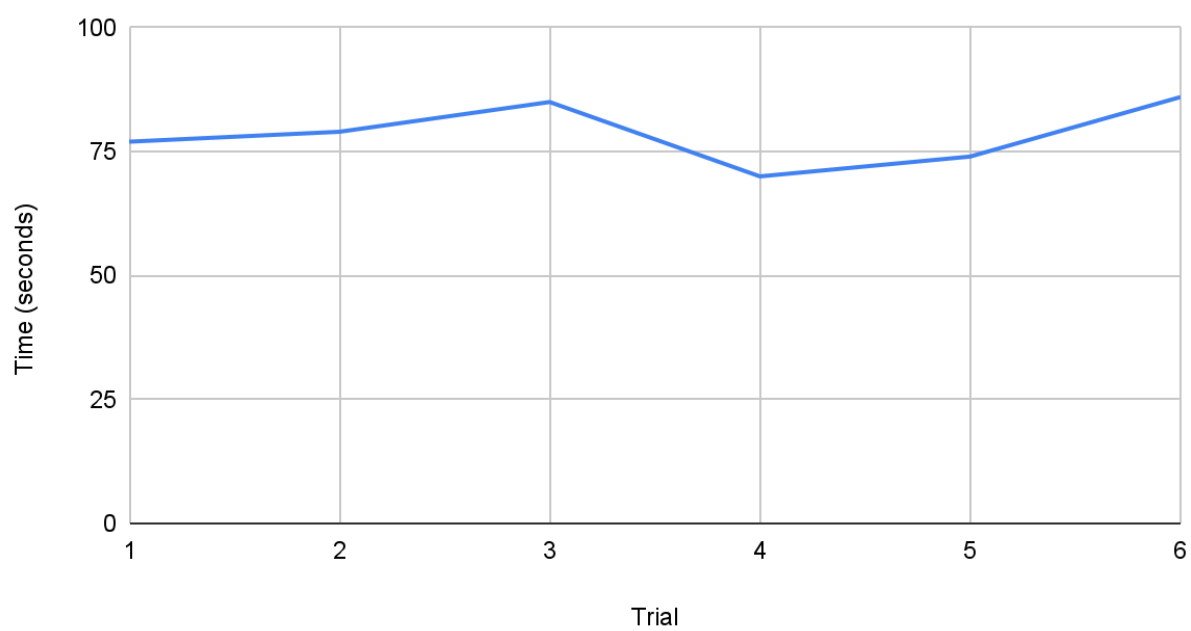


Figure 7 - Graph of Voltage Test after 2 Course Loops

Voltage Over 1 Hour After 2 Course Traversals

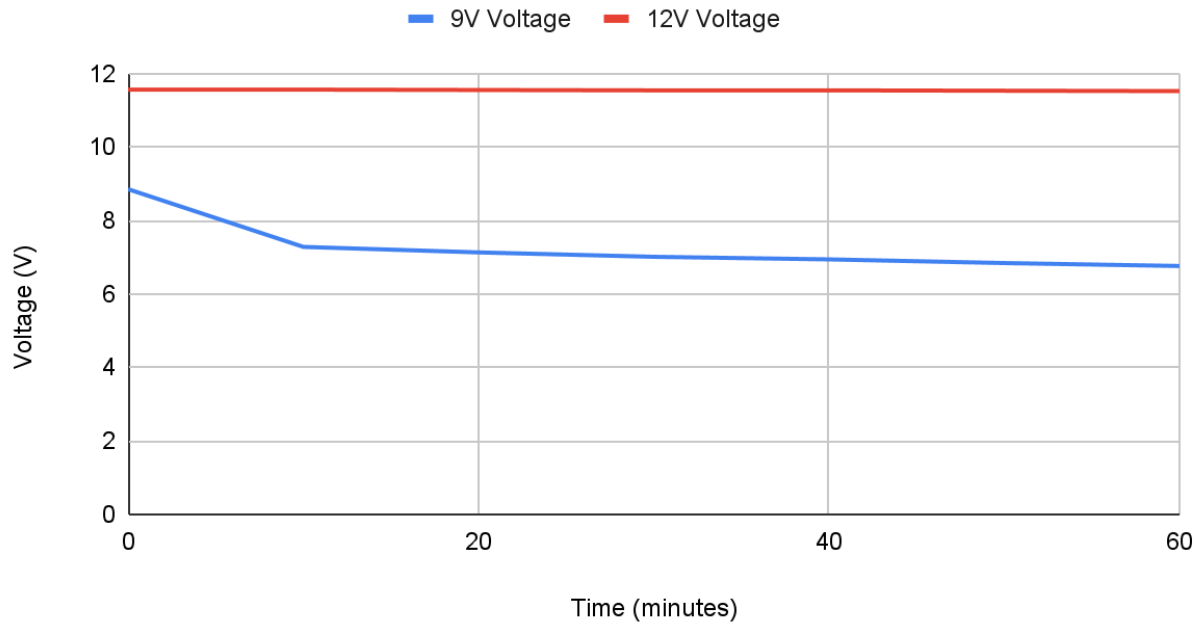
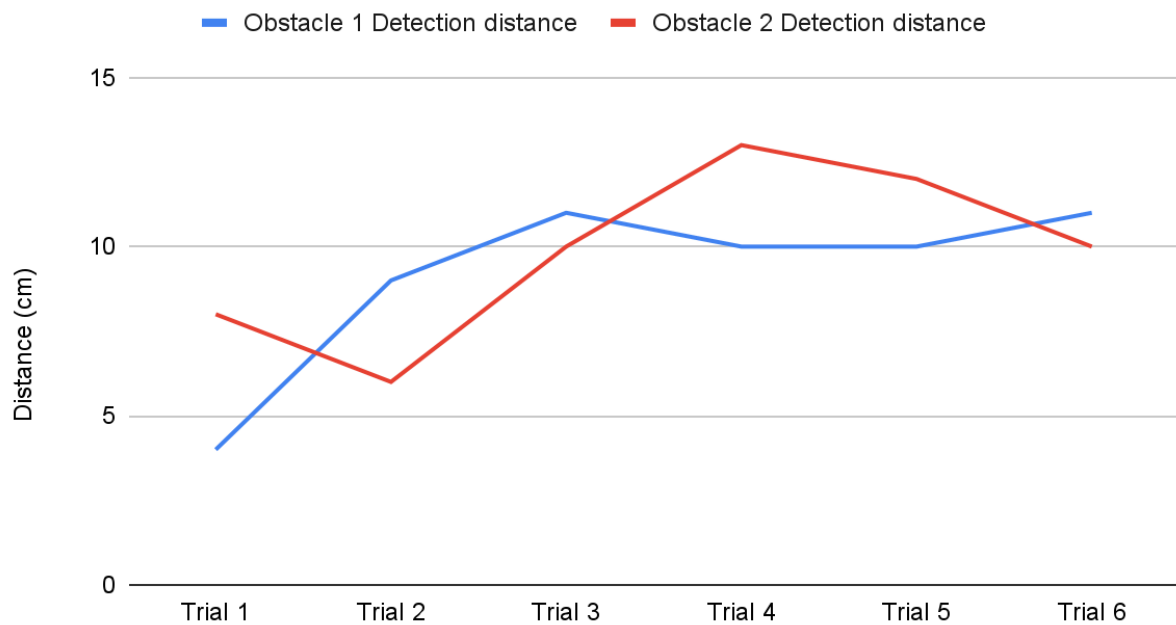


Figure 8 - Graph of Obstacle Detection Distances Over 6 Course Traversals

Obstacle Detection During Line-Following



Appendix B - Tables

Table 1 - Pairwise Comparison chart (PCC)

Goals	Accurate Path Tracking	Cost efficiency	Speed	Durability	Simplicity	Portability	Short response time	Score
Accurate Path Tracking	-	1	1	1	1	1	1	6
Cost efficiency	0	-	1	1	1	1	1	4
Speed	0	0	-	0	1	1	0	2
Durability	0	0	1	-	1	1	1	3
Simplicity	0	0	0	0	-	1	0	1
Portability	0	0	0	0	0	-	0	0
Short response time	0	0	1	0	1	1	-	3

Table 2 - Morph Chart

Design 1 Design 2	1	2	3	4	5	6	7
Move	Wheels	Treads	Robotic legs	Hover	-----	----- -	----- -
Detect line	Color sensor	Standard camera	Monochrome camera + edge detection	Lidar sensor	Infrared sensor	----- -	----- -
Processing/ controller	Arduino Uno	Arduino Nano	Raspberry pi pico	ESP32	----- -	----- -	----- -
Detect object	Color sensor	Standard camera	Monochrome camera	Lidar sensor	Infrared sensor	Ultrasonic sensor	Stereo cameras

Detect finish line	Color sensor	Standard camera	Monochrome camera + edge detection	Lidar sensor	----- -	----- -	----- -
Touchless activation	Bluetooth receiver	Wifi module	Radio frequency module	Infrared module + remote	Camera, activate based on visual hand movements	----- -	----- -
Run for 1 hour	Battery	Diesel engine	Hydrogen fuel cells	----- -	----- -	----- -	----- -
Actuate wheels/treads	Stepper motor	DC Motor	AC Motor	360 Servo motor	----- -	----- -	----- -
Steering	Servos	Stepper motor	DC Motors w/gearbox	Brushless DC motor	Linear actuator + gears	----- -	----- -
Contain components /carry load	Steel chasis	Aluminum chasis	Sheet metal chasis	FDM Chasis	Resin printed	Resin + metal hybrid	FDM + metal hybrid

Table 3 - Bill of Materials

Bill of Materials	Number	Cost per Unit (USD)	Total Cost (USD)
12V Li-Ion Battery	1	32.99	32.99
9V Battery	1	2.5	2.5
3-6V DC Motors	4 (pack of two + driver)	7.99	15.98
Motor Drivers	2	See above	N/A
Infrared Sensor	3	0.88	2.64
Remote Signal Receiver	1	1.95	1.95
IR Remote Control	1	4.95	4.95
1:48 Wheels	4	0.99	3.94
Arduino Uno Board	1	8.99	8.99
		Final Component Cost (USD):	73.94

Table 4 - Results of Course Traversals

Course test			2 per trial	
Trial	Time (seconds)	Successful? (Y/N)	Obstacle detection rate	
1	77	Y	100%	Avg time:
2	79	Y	100%	78.5
3	85	Y	100%	
4	70	Y	100%	
5	74	Y	100%	
6	86	Y	100%	

Table 5 - Idle Power Test Data

Idle Power test (after 2 course tests)		
Time (minutes)	9V Voltage	12V Voltage
0	8.85	11.56
10	7.28	11.56
20	7.13	11.55
30	7.01	11.54
40	6.94	11.54
50	6.84	11.53
60	6.76	11.52

Table 6 - Course Traversal Obstacle Detection Distances

	Trial 1	Trial 2	Trial 3	Trial 4	Trial 5	Trial 6
Obstacle 1 Detection distance (cm)	4	9	11	10	10	11
Obstacle 2 Detection distance (cm)	8	6	10	13	12	10

Appendix C - Objectives/Metrics

- Accurate Path Tracking - Some part of the car must remain on the line at all times
 - Metric: Does at least some part of the chassis of the vehicle remain above the line throughout the entire path?
- Cost Efficiency
 - Metric: Is the final vehicle bill of materials within the \$200 budget?
- Speed
 - Metric: Does the vehicle complete the track in 90 seconds?
- Durability - structure can withstand moderate stress, doesn't collapse under its weight and all parts remain secure during operation
 - Metric: Does the vehicle withstand a safety margin of 750g additional weight well enough to avoid structural failure and move as intended?
- Short response time - quickly senses and responds to changes in the environment. Must stop within 10 cm of an object, after getting stopped by an object must start moving in less than 3 seconds.
 - Metric: Does the front of the vehicle stop 10 cm in front of an object? Do its movement mechanisms begin to activate the instant after an obstacle is removed?
- Portability -
 - Does the vehicle fit in length and width upon an 8.5x 11" sheet of paper

Constraints:

- Tetherless
 - Metric: Is the vehicle powered by a source inside/directly fixed atop its chassis?
- Indicate when crossing the finish line
 - Metric: Some difference in physical appearance or audio signal noticeable from at least 20 feet away (depending on the length of the course, the distance from the beginning of the course to the end of the course + 10 feet for spectator viewing)
- Chassis cannot be too simple or overly complicated (too many cuts and edges)
 - Metric: Is the chassis deemed too simple or overly complicated by the client?
- Size roughly around 8.5x11" sheet of paper
 - Metric: Able to fit within an 8.5x11" area when viewed from the top-down
- Able to carry 250g load
 - Metric: Can carry 250g additional load and still complete the path/adhere to the other constraints
- Must be operated without touching the vehicle itself
 - Metric: Operator does not physically touch vehicle to activate it
- Runtime of 1 hour
 - Metric: Vehicle has battery capacity and power draw allowing for 1 hour of constant motion

- Recognize different color finish line
 - Metric: Vehicle stop code runs when the sensor detects the finish line
- Stop 10 cm from obstacle
 - Metric: Front of vehicle stops within 10 cm of an obstacle
- Respond to obstacle being removed within 3 seconds
 - Metric: The vehicle takes no longer than 3 seconds to begin moving after an obstacle is removed
- Turn 135 degrees or more
 - Metric: Vehicle successfully (abides by all constraints) completes practice course with 135+ degree turns
- Vehicle steer must be powered by some sort of actuator
 - Metric: Vehicle steer is powered by some sort of actuator
- Must finish the course in under 2 minutes
 - Metric: Vehicle finishes the course in under 2 minutes

Appendix D - Power Calculations

Current consumption of different components (at maximum sustainable current draw not including stall):

Obstacle IR sensor = 20 mA

Arduino Uno = 50 mA

Remote Receiver IR sensor = .45 mA

DC Motor = 160 mA

L298 Motor Driver = 36 mA

Based on the following equation, the total draw can be determined from current:

Eq(1)

$$\text{Battery Charging Time (in Hours)} = \frac{\text{Battery Capacity (in mAh)}}{\text{Charging Current (in mA)}}$$

Total draw on arduino power system: $3 * 20 + 50 + .45 = 110.45 \text{ mA}$

- Arduino power source = 9V battery = 550 mAh

Total draw on motor system: $4 * 160 + 2 * 36 = 712 \text{ mA}$

- Motor power source = 12V battery = 6000 mAh

Therefore, the 9V battery will last $550/110.45 = 4.98$ hours and the 12V power supply will last $6000/712 = 8.427$ hours.

Appendix E - References

[1]“Arduino DC Motor Control,” Tutorialspoint [Online]. Available:

https://www.tutorialspoint.com/arduino/arduino_dc_motor.htm. [Accessed: 30-Apr-2025].

[2]“Obstacle detector using IR module - tutorial,” Arduino Project Hub [Online].

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